1. Alexnet(David Ojo/801126859/HW1/ https://github.com/David-Ojo/UNCC-ECGR-4106)

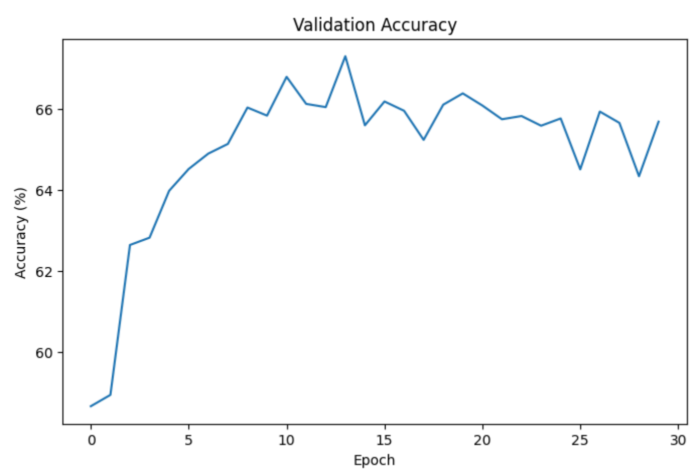
A.

Initially tried to run the model but ran into an error of "`Output size is too small` " which makes sense due to the 227x227 going down to a 32x32.

So looking at the layers are shaved down:
32×32
→ 7×7
→ 3×3
→ 3×3
→ 1×1
→ 1×1
→ 1×1
→ 1×1
→ ERROR
It can be seen that Alexnet in it's base form crushes any 32x32 put into it. And by the 12th layer it gives an error

The first improvement is making it so that the Alexnet doesn't give an error. Since the final layer appears to make the output too high, I removed the third max-pooling and replaced the original classifier as it was far too large for 32x32. I also removed dropout so i can compare it to when I add it back in during section b.

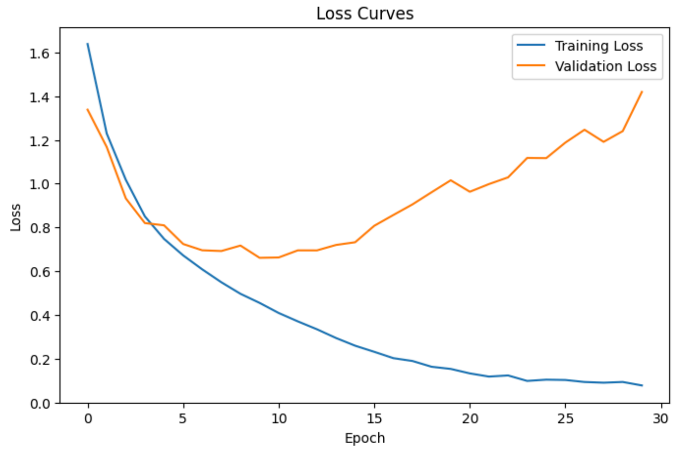
Looking at the Initial graphs:

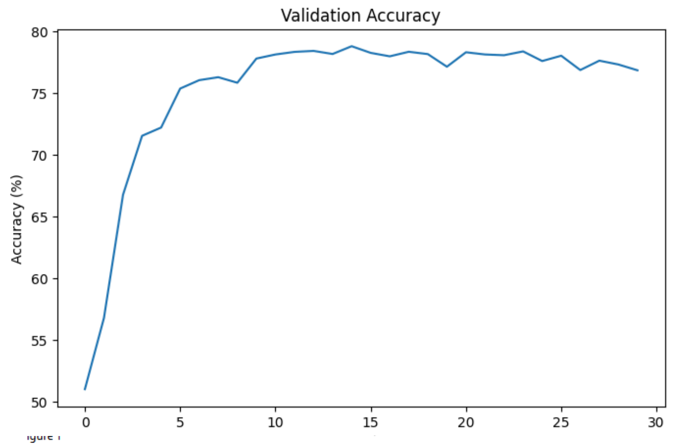


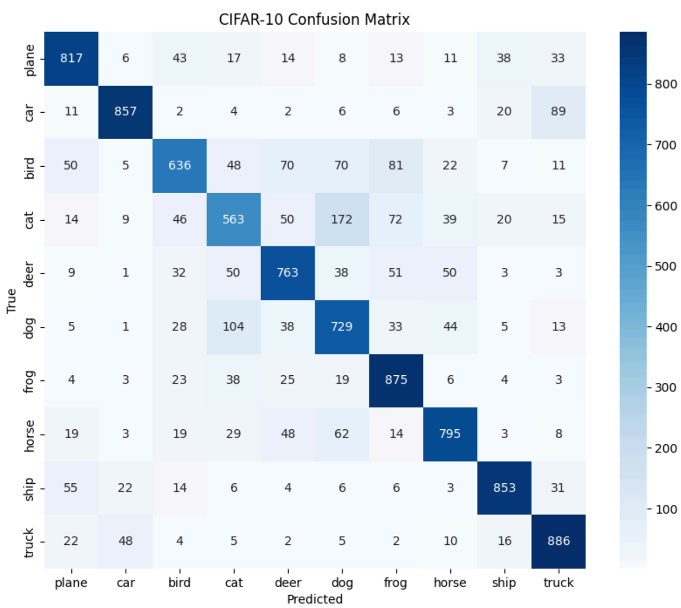
Overfitting can be clearly seen from the validation loss diverging from the training loss.

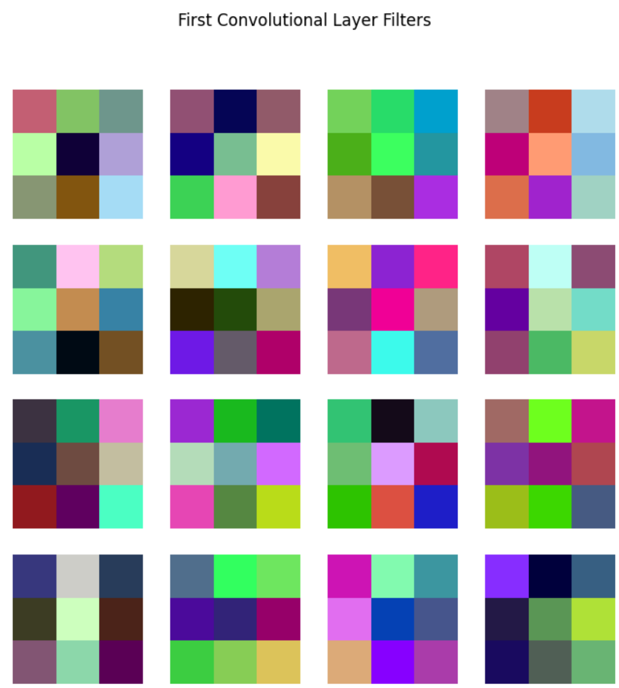
To address this I reduced the kernel which was 11x11 stride 4 to 3x3 stride 1 to better fit 32 x 32, I also reduced the filer layers from 64, 192, 384, 265, 265 to 32-64-64-128-128. Lastly instead of

calling Alexnet inside collab I am using a fully connected layer that follows the same structure as Alexnet but is far smaller so as to better fit a 32x32 dataset.
Doing so produces:

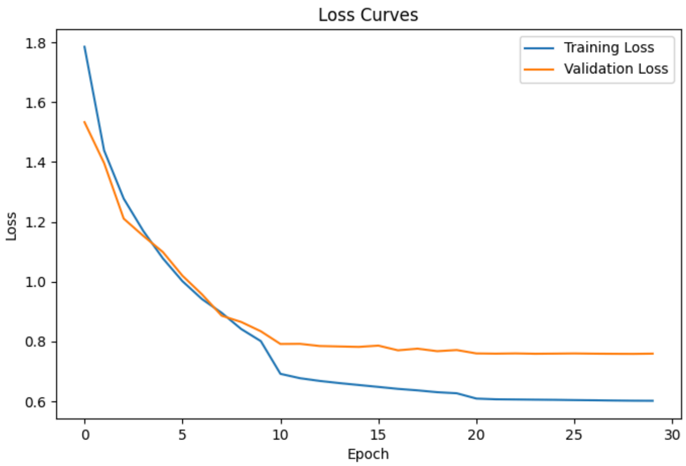


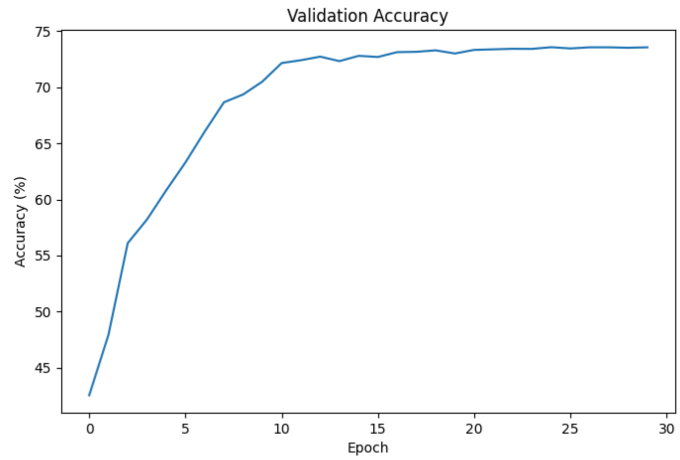


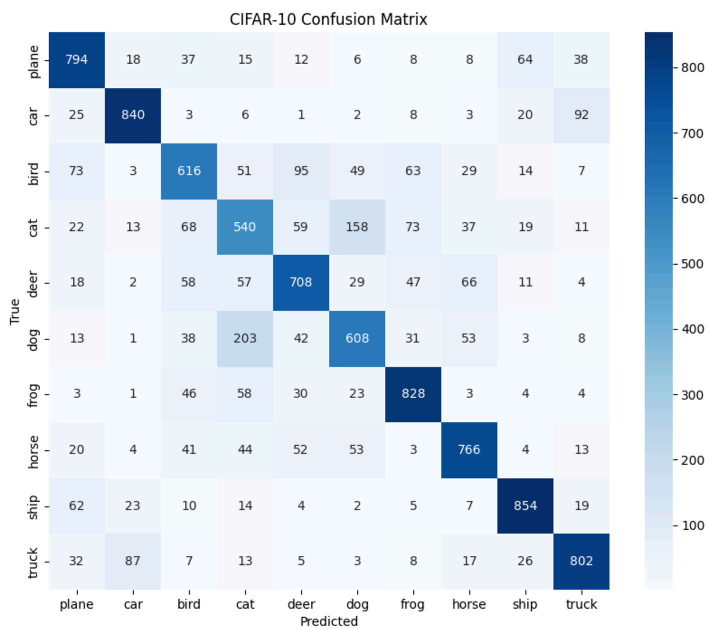


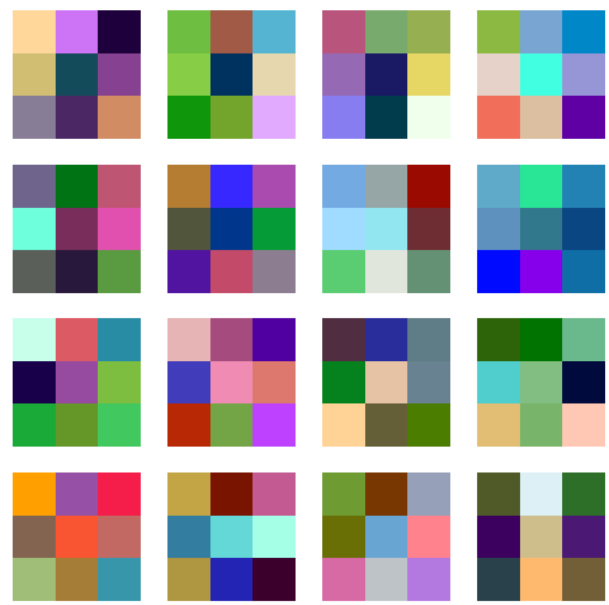


Which, while still overfitting, is an improvement of over 10 percent accuracy and significantly lower validation loss at the start. After this Collab did not let me use the gpu for over 24 hours so I moved to my personal setup which has a GTX Titan X (Pascal) and an AMD Ryzen Threadripper CPU . I added a schedular and weight decay to the classifier, as well as making the model even more simple. I considered adding data augmentation but decided against it as the instructions say “ simplify the model” and that isn’t model simplifying.





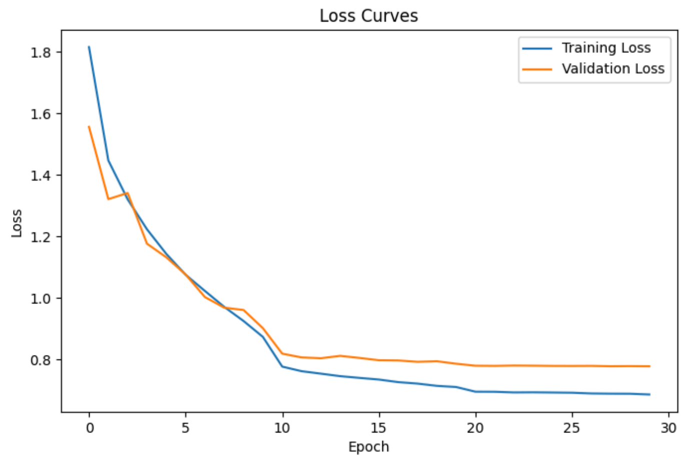


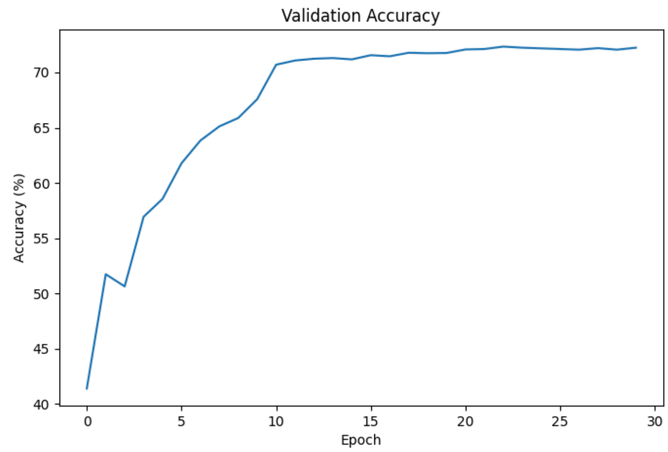


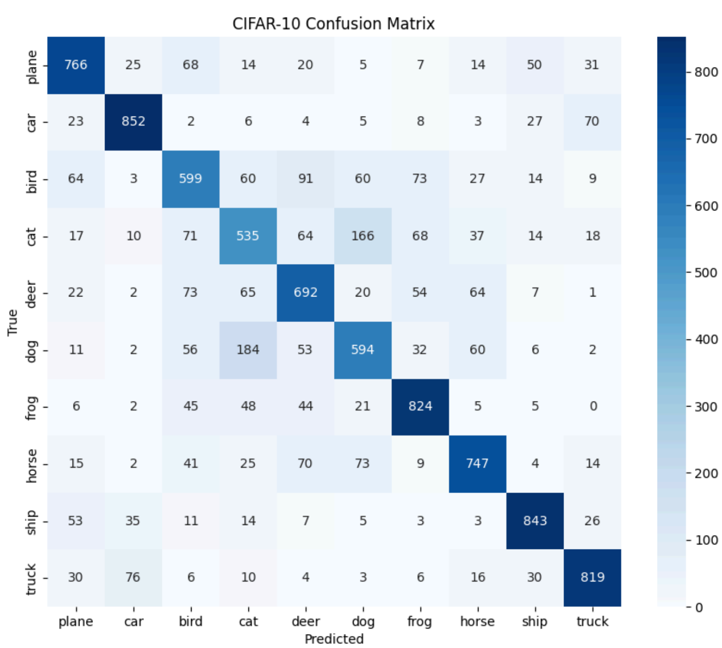
Epoch [30/30] Train Loss: 0.6021 Test Loss: 0.7596 Test Acc: 73.56%
Trainable Parameters: 79,370

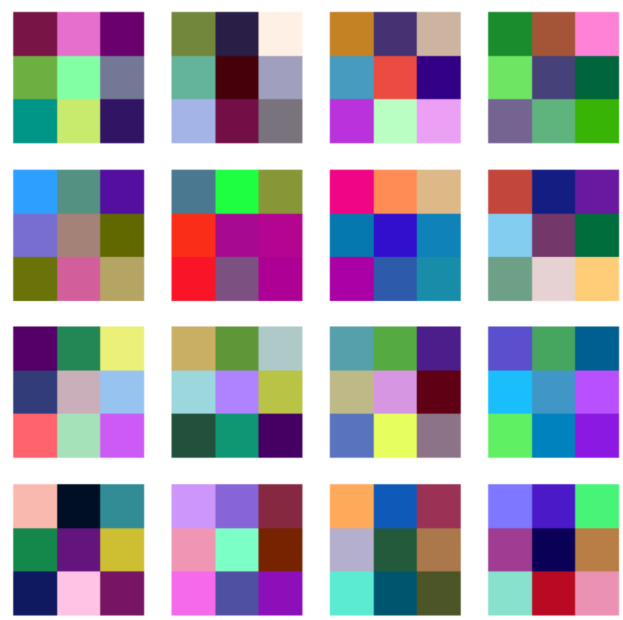
B. Introducing DropOut

I placed a dropout of p = .5 in my classifier after my Relu.



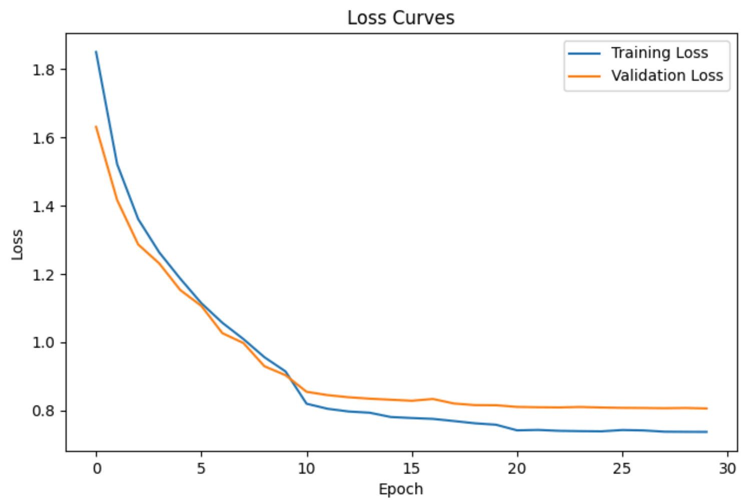


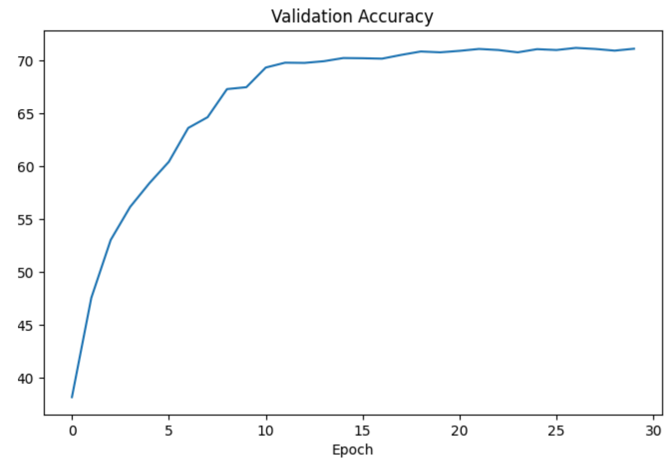


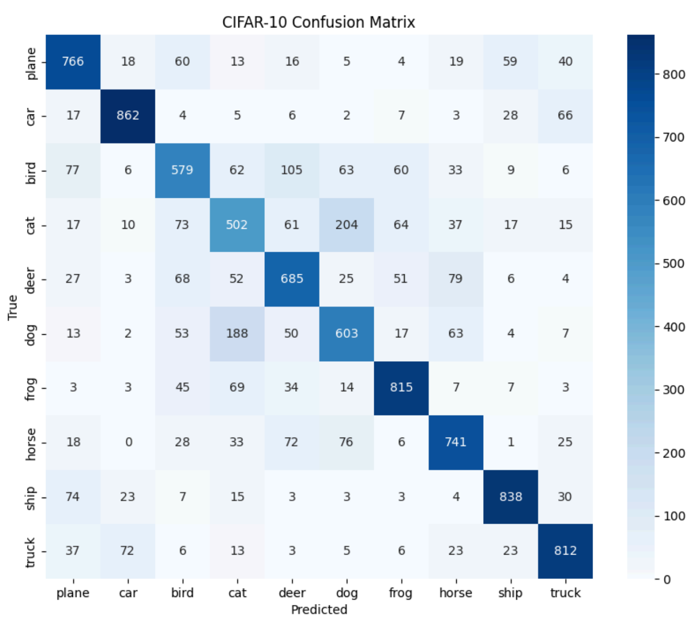


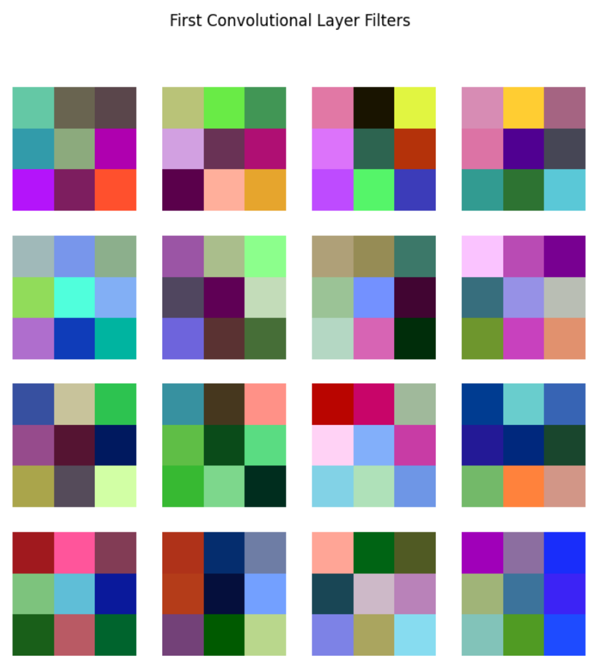
Epoch [30/30] Train Loss: 0.6866 Test Loss: 0.7897 Test Acc: 72.71%
Trainable Parameters: 79,370

As well as p = .3









Epoch [30/30] Train Loss: 0.7376 Test Loss: 0.8111 Test Acc: 72.03%
Trainable Parameters: 79,370

Introducing dropout Reduced performance when compared to the non dropout model but it seemingly reduced the gap between the val and train loss.

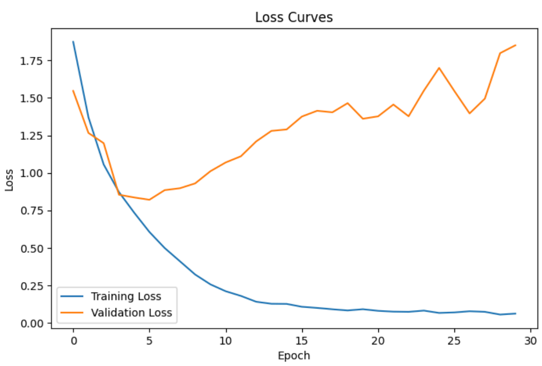
A dropout rate of 0.3 provided the worst performance as it has the lowest accuracy and the highest loss of the three.

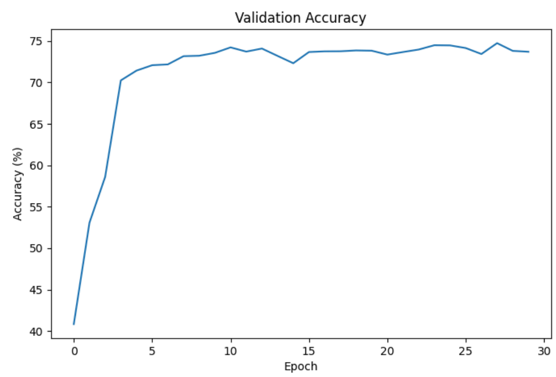
The total number of parameters for this model is : Trainable Parameters: 313,354
Compared to Alexnet's : 61,400.840

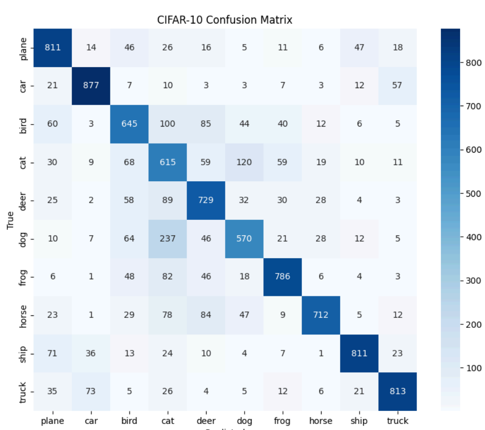
I don't notice much of a pattern when visualizing the Layers of the model.

2. VGG

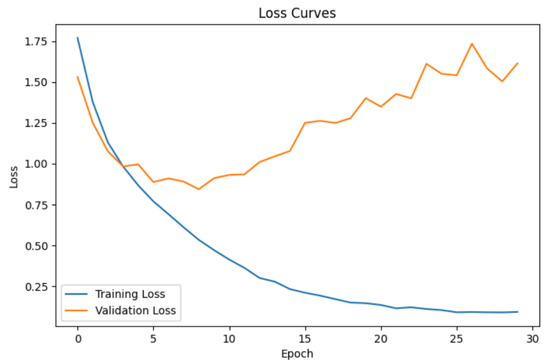
A. Since VGG-11 is smaller I opted to use that as when calculating its parameters of 9,488,266 dwarfs the parameter count of my simplified Alexnet.
Like Alexnet, VGG-11 straight won't run so I turned it into a sequential equivalent of the repeated convolution layers.

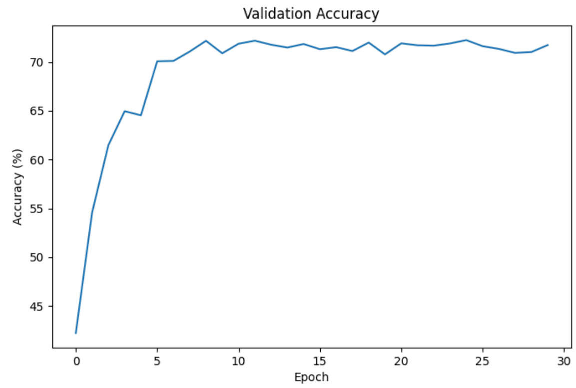






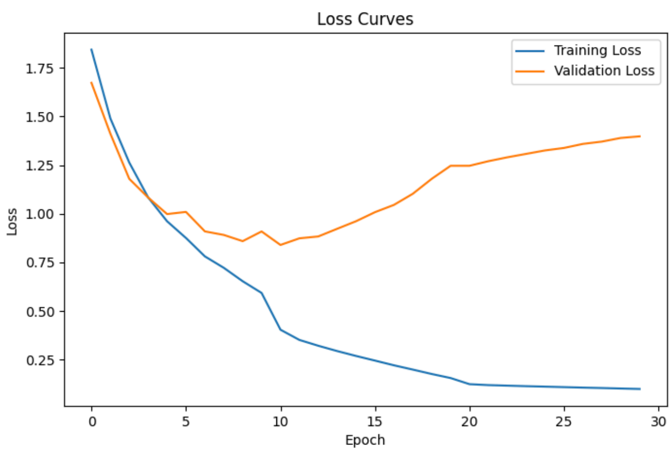
Looking at the initial 9.5 mill parameter it is very close to the modified Alexnet.
In order to try and keep this level of accuracy while reducing the parameter count I first reduced the channel count of all of the layers By around 3/4ths keeping the number of convolutions and poolings just making the model overall leaner.

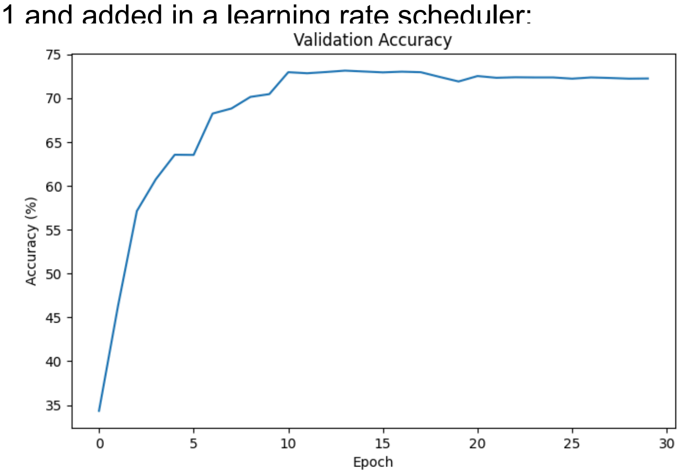


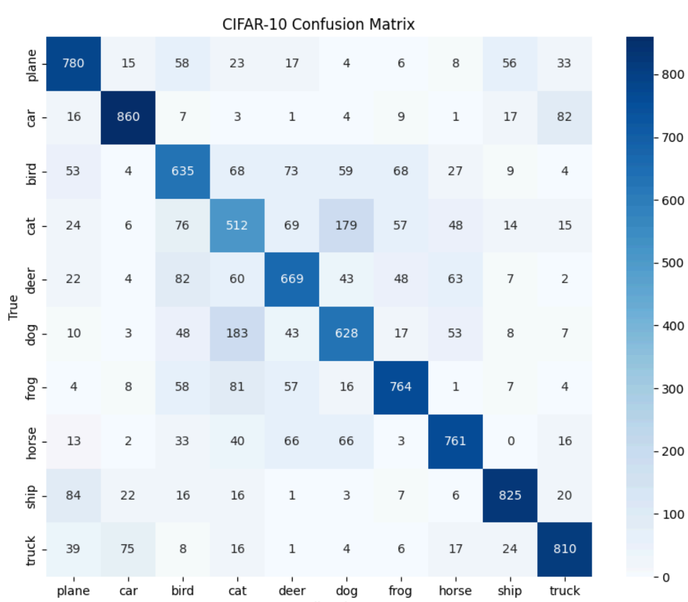


When reducing the channel count it results in “Trainable Parameters: 612,714” which is over a 90% decrease in parameter count while only losing around a single percentage of accuracy. It also trained much faster than the full version.

Since it was still overfitting I did what I did for problem 1 and added in a learning rate scheduler:



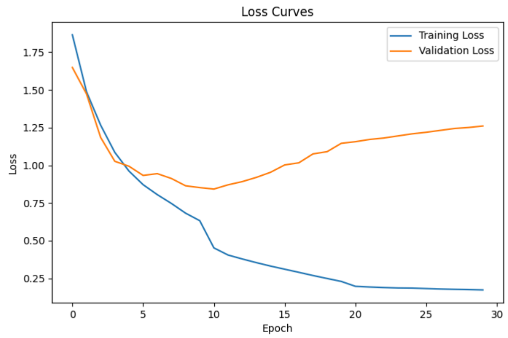


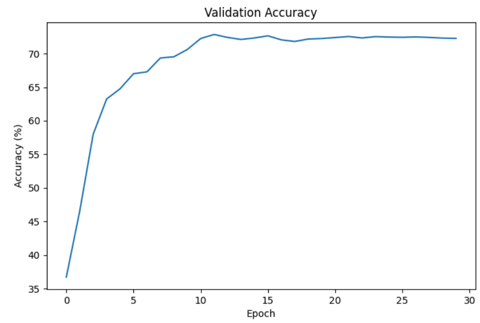


Epoch [30/30] Train Loss: 0.0995 Test Loss: 1.4577 Test Acc: 72.44%
Trainable Parameters: 612,714
This made the process more stable but did not do much to correct overfitting but it did increase the accuracy.

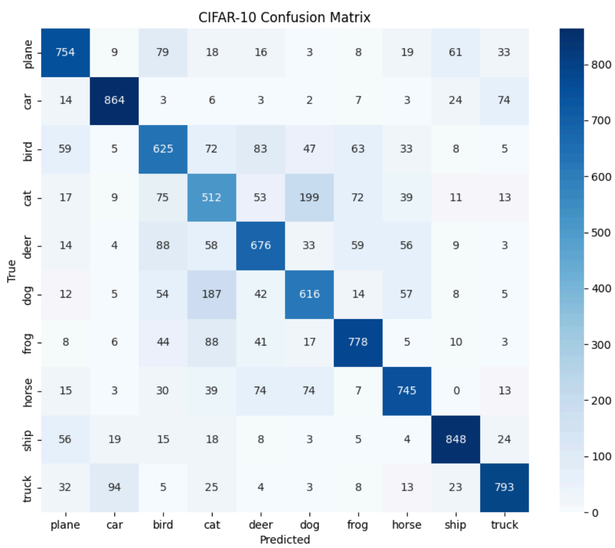
B.Introducing Dropout

For Dropout I initially used p = .3 which results in:



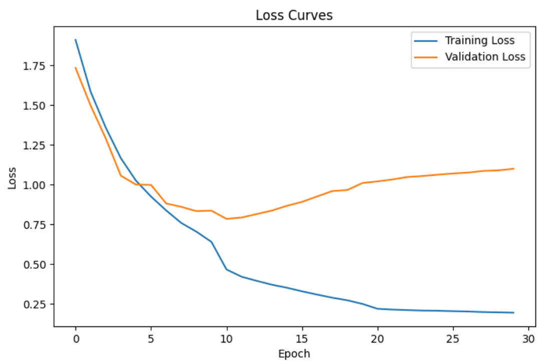


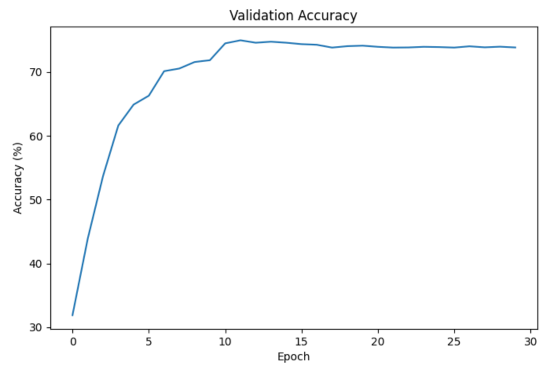
Epoch [30/30] Train Loss: 0.1730 Test Loss: 1.2963 Test Acc: 72.11%
Trainable Parameters: 612,714

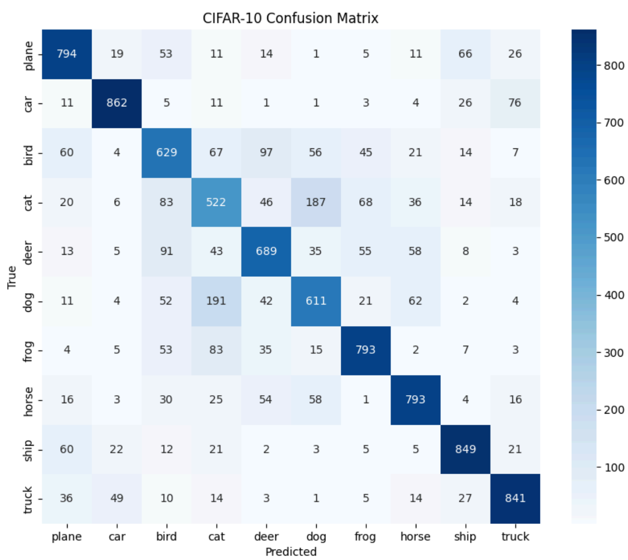


Epoch [30/30] Train Loss: 0.1730
Test Loss: 1.2963 Test Acc: 72.11%
Trainable Parameters: 612,714

And p = .5







Epoch [30/30] LR: 0.000001 Train Loss: 0.1947 Val Loss: 1.0998 Val Acc: 73.84%
Epoch [30/30] Train Loss: 0.1947 Test Loss: 1.1219 Test Acc: 73.83%

When looking at non Drop and the model that used Dropout .5 had the lowest val loss at .2111 compared to the non dropout that had .2499. Drop out of .3 has the highest val and training loss and the lowest accuracy of the three. This contrasts with Alexnet whose best performance was with no dropout at all.

Model	No Dropout	Dropout 0.3	Dropout 0.5	Best
AlexNet	~74%	Lower	Lower	No Dropout
VGG-11	Lower	Lower	Best (~74%)	Dropout 0.5

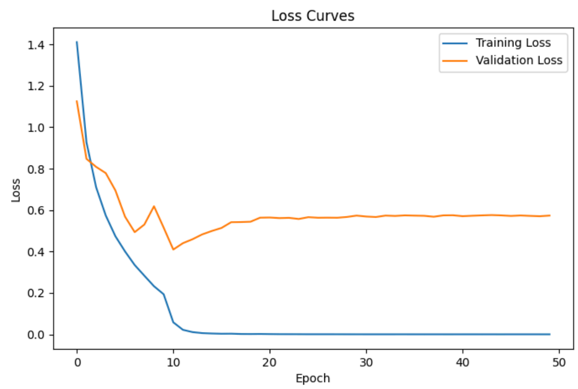
Since VGG is a deeper model which is shown by the higher number of parameters it makes sense that dropout has a more positive effect. When referring to the best models of each while Alexnet was 1 percent more accurate, VGG was nearly just as accuracy in 13 less epochs and with each epoch taking less time. This can probably be attributed to the VGG model being deeper and probably overfitting to a certain degree.

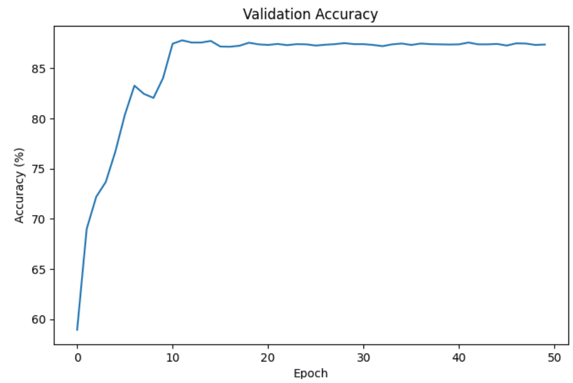
3. Resnet

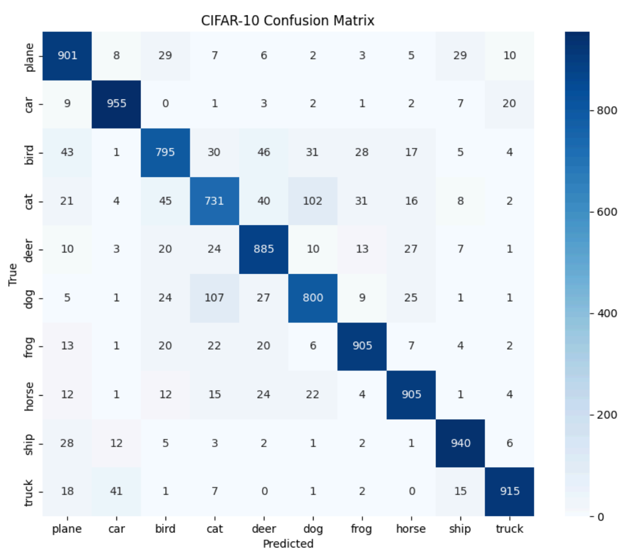
There is no Resnet from the lecture. I created what I think ResNet 11 would be, which is a simpler version of Resnet 18.

A.

For Resnet 18:

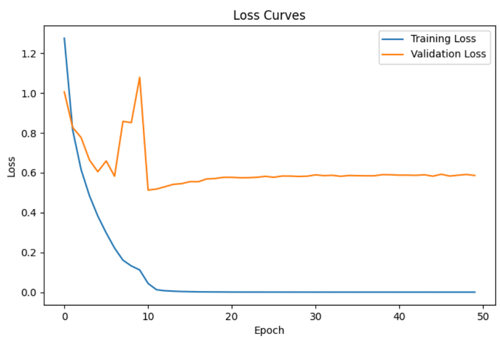


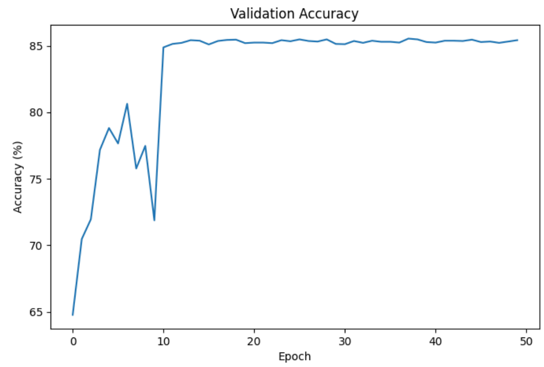


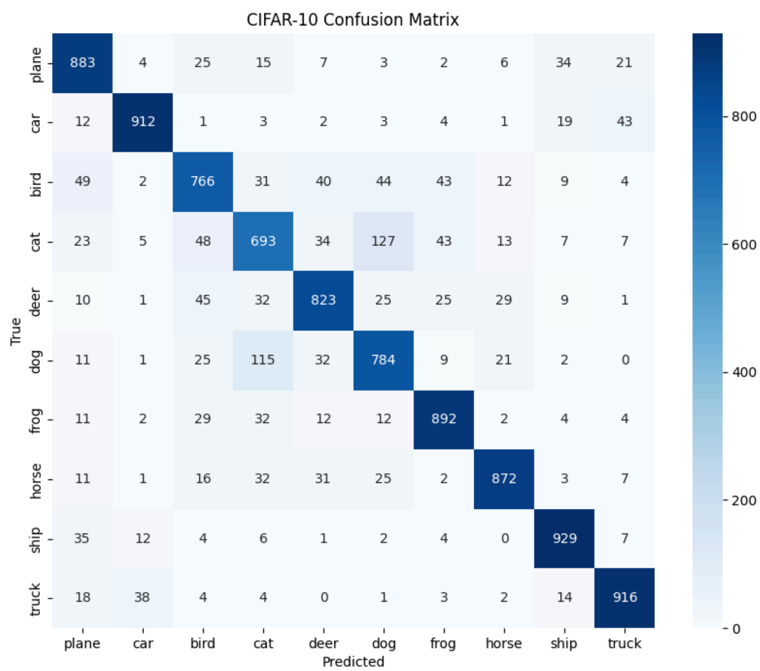


Epoch [50/50] Train Loss: 0.0005 Test Loss: 0.5929 Test Acc: 87.32%
Trainable Parameters: 11,173,962

For Resnet 11:







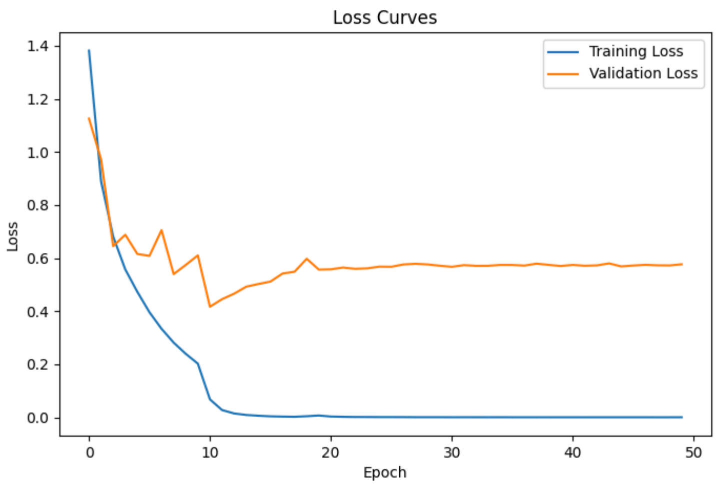
Epoch [50/50] Train Loss: 0.0006 Test Loss: 0.6622 Test Acc: 84.70%
Trainable Parameters: 4,903,242

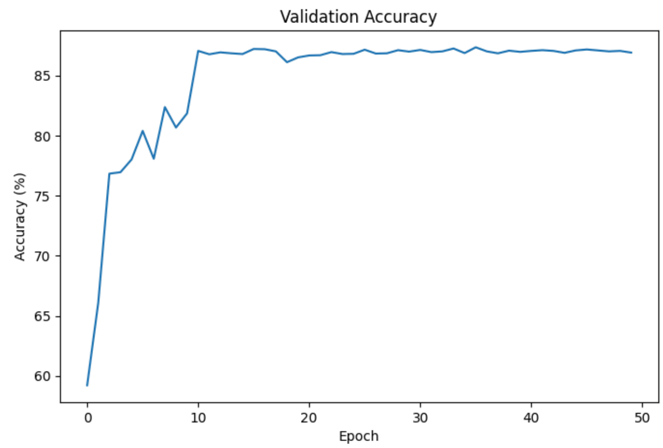
Looking at the table Resnet 11 provides enough performance for less than half the parameters

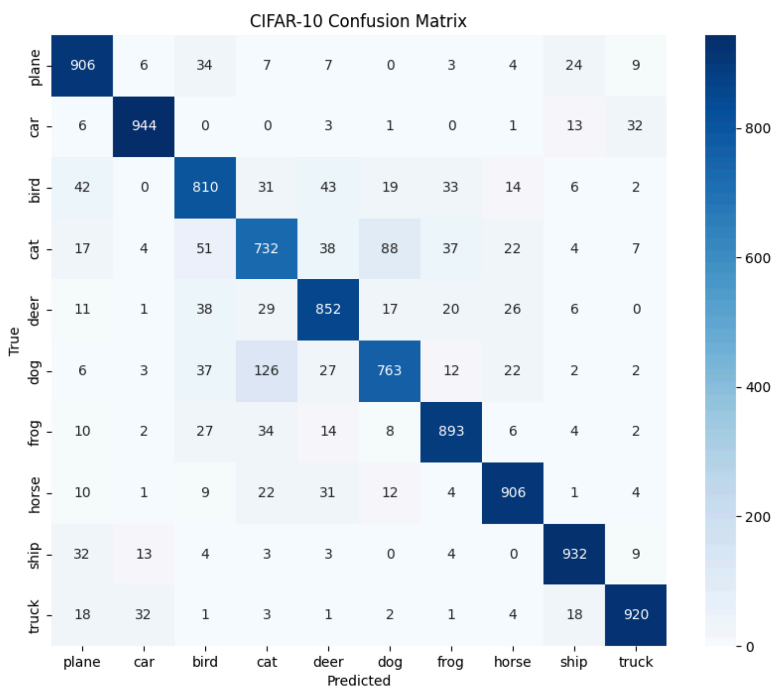
Model	**Parm**	**Acc**	**Time per epoch**
Resnet- 18	11,173,962	87.32%	61.06 sec
Resnet- 11	4,903,242	84.70%	39.06 sec

B. Introducing Dropout

Dropout .03 - Resnet 18



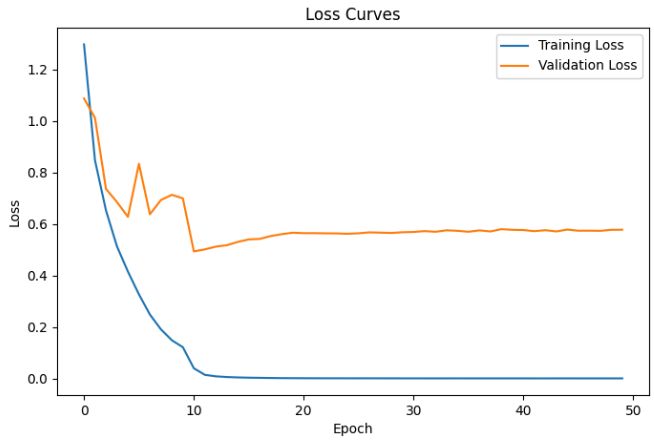


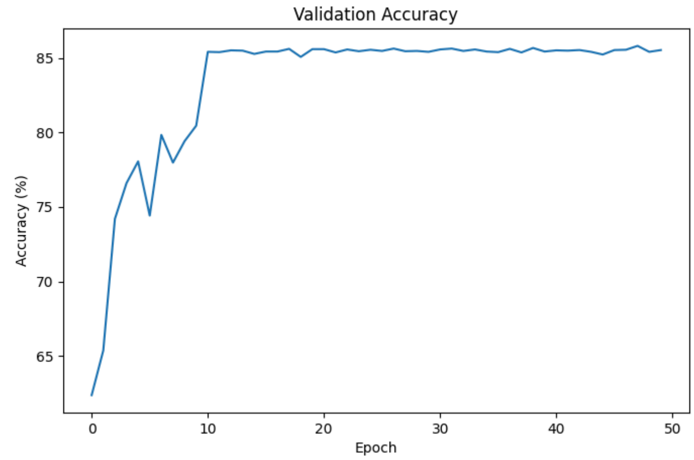


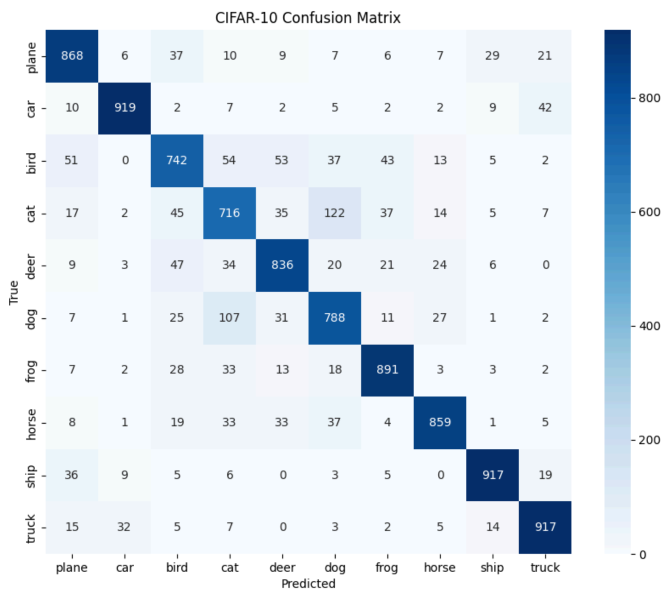
Epoch [50/50] Train Loss: 0.0008 Test Loss: 0.6145 Test Acc: 86.58%

Trainable Parameters: 11,173,962

Resnet 11 -

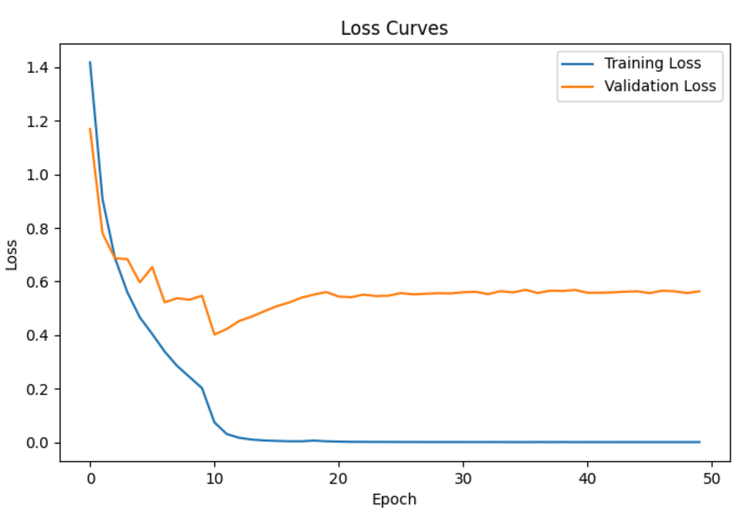


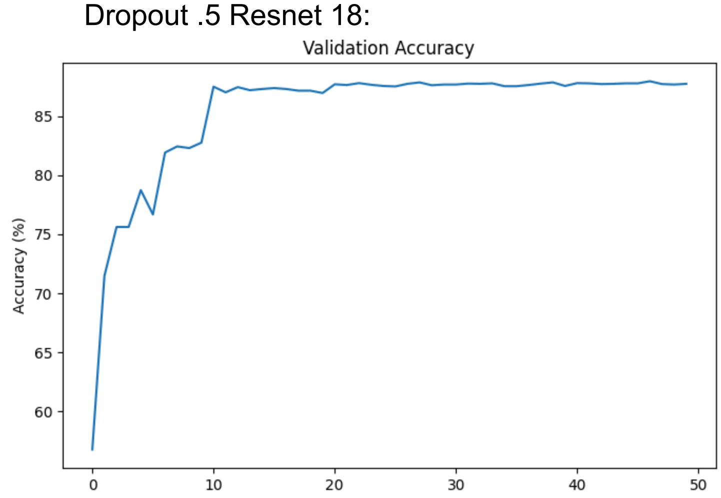




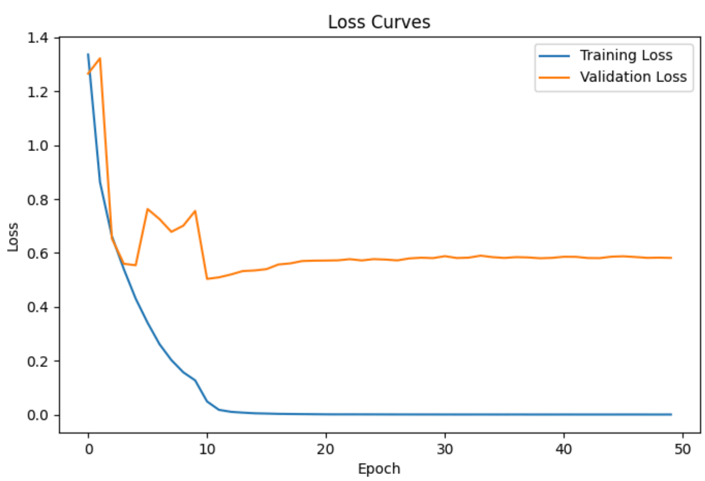
Epoch [50/50] Train Loss: 0.0008 Test Loss: 0.6397 Test Acc: 84.53%
Trainable Parameters: 4,903,242

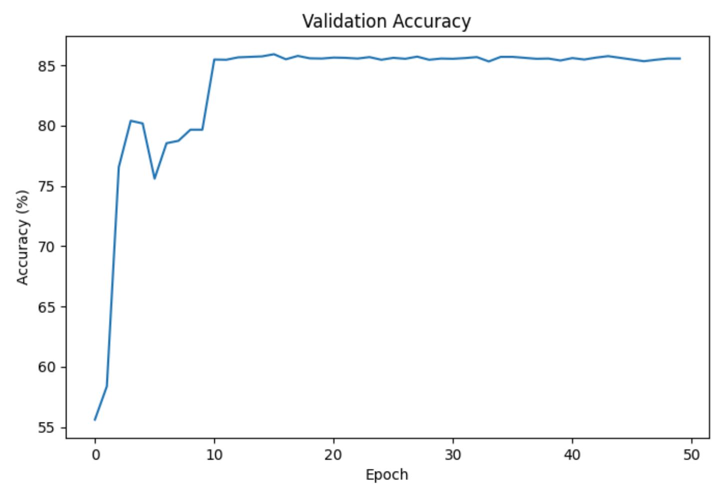
Dropout .5 Resnet 18:

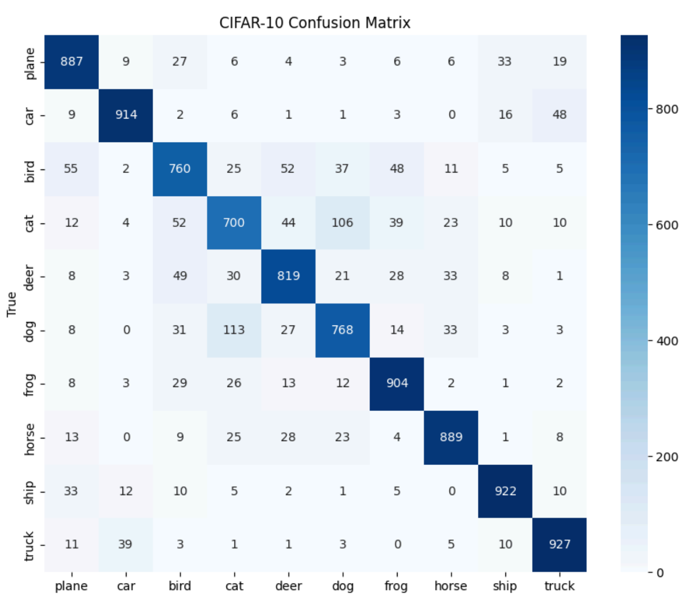




Resnet 11:







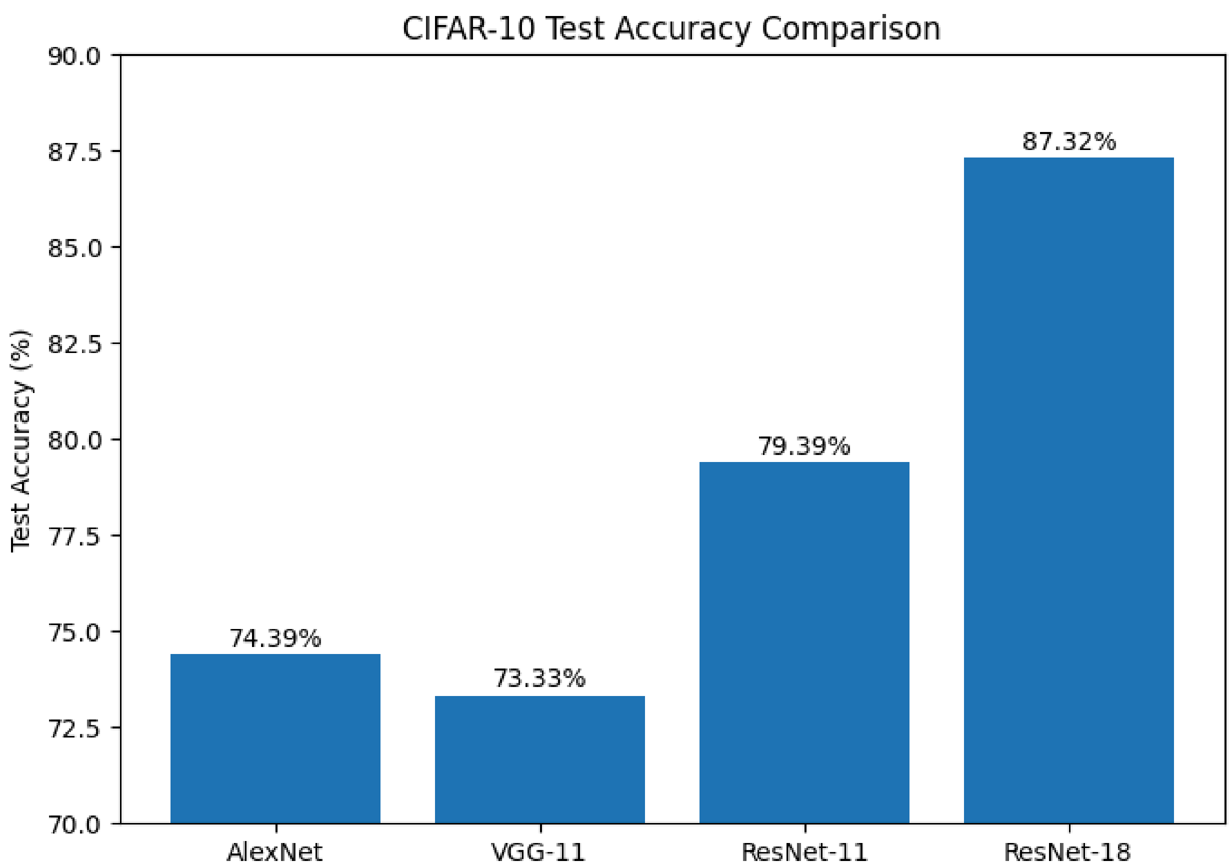
Epoch [50/50] Train Loss: 0.0011 Test Loss: 0.6342 Test Acc: 84.90%
Trainable Parameters: 4,903,242

Like VGG and unlike Alexnet introducing Dropnet improved performance with both Resnet models seeing the biggest boost at Dropnet = .5

Best Models:

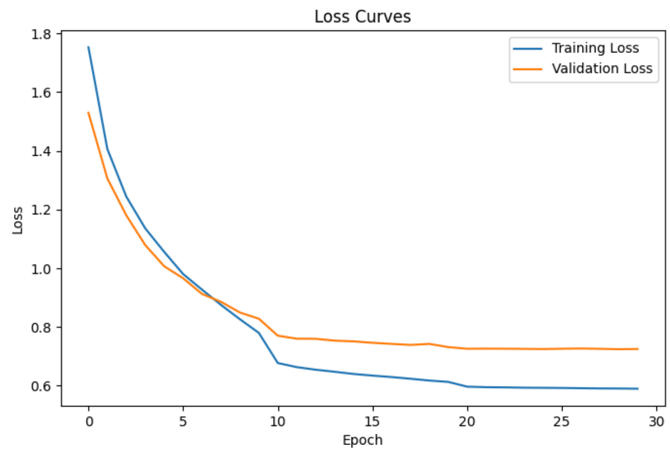
For this section I used a parameter sweep style where I went through a multitude of different seeds to search for the best. Early stop was employed where 5 epochs go without improvements it will go to the next seed.

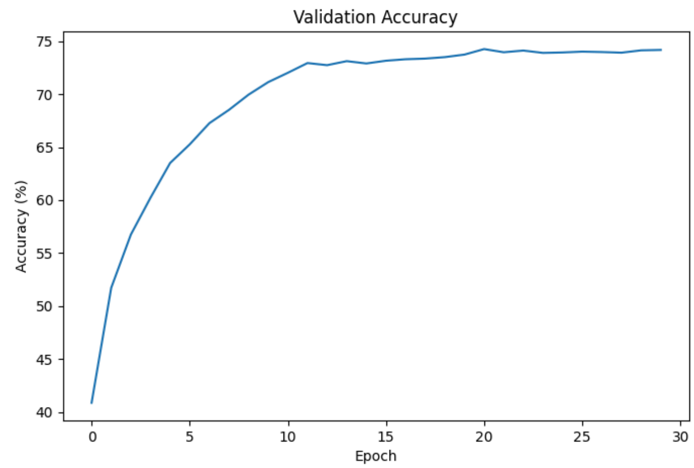
Model	Parm	Acc	Time per epoch
AlexNet	79,370	74.39%	15.41 sec
VGG-11	612,714	73.33%	14.63 sec
Resnet- 11	4,903,242	79.39%	39.06 sec
Resnet- 18	11,173,962	87.32%	61.06 sec

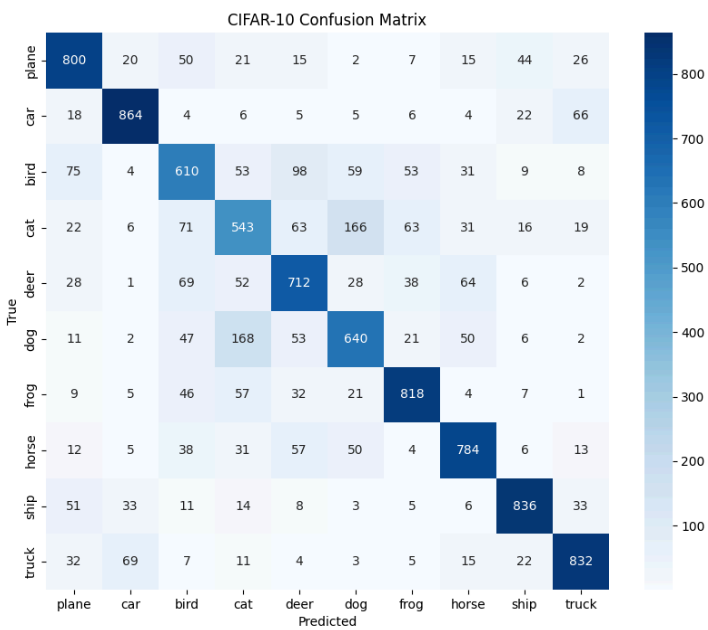


As can be seen, there is a mostly direct like with the complexity of the model and it's performance with VGG11 being the outliner there is a direct line of improvement from parameter count but when taking into consideration how much smaller Alexnet it, I wouldn't say it has no place.

Alexnet:(No dropout as that was one best one in initial testing)

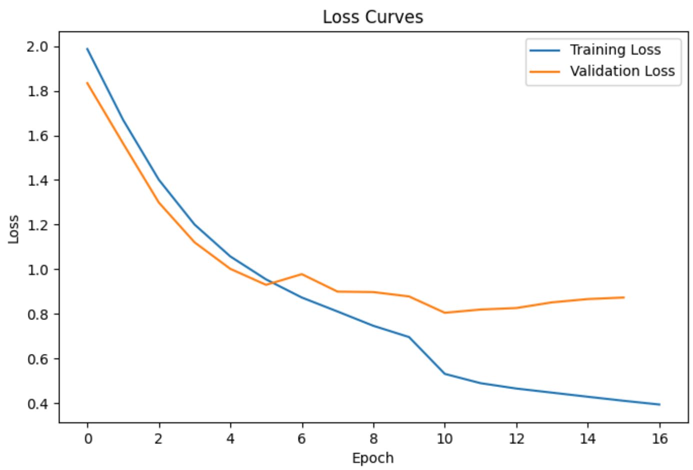


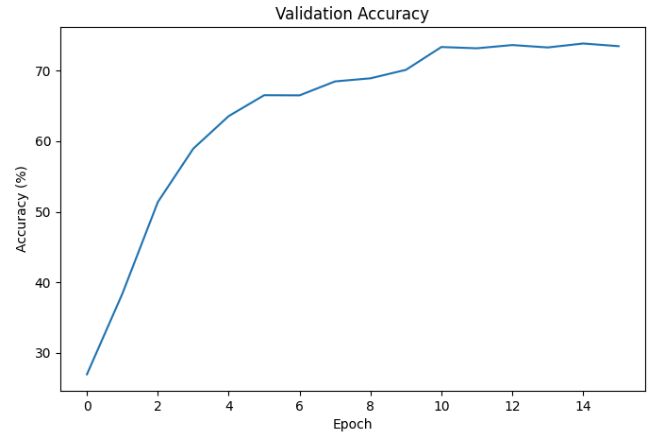


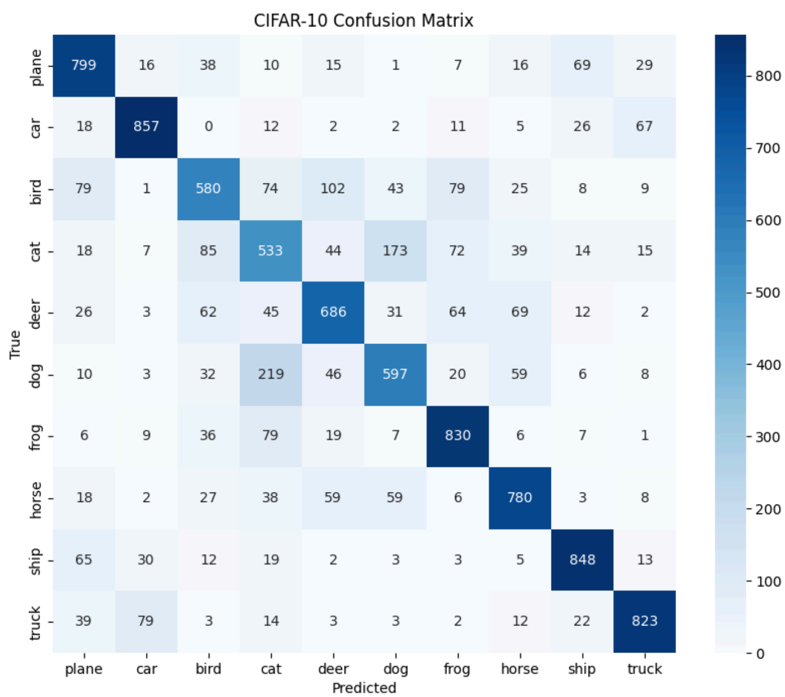


Total Training Time: 462.38 sec
Average Epoch Time: 15.41 sec
Seed 1: 74.39%
Epoch [30/30] Train Loss: 0.5900
Test Loss: 0.7478 Test Acc:
74.39%
Trainable Parameters: 79,370

VGG-11(Dropout = .5 was used as it performed the best)

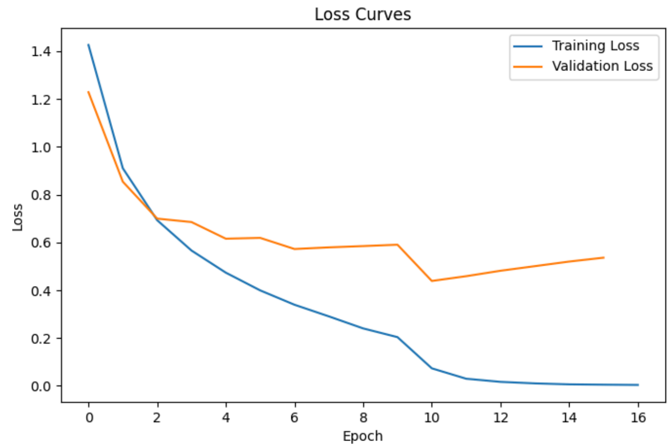


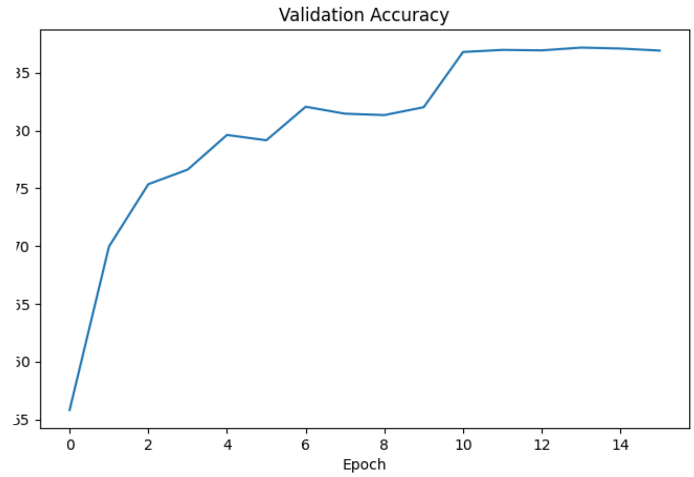


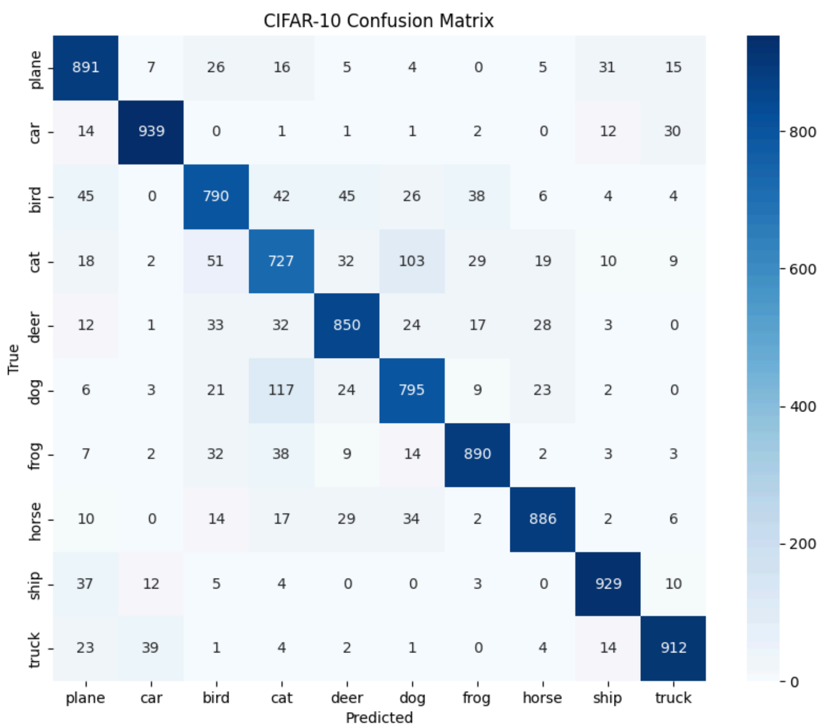


Total Training Time: 247.57 sec
Average Epoch Time: 14.63 sec
Seed 2: 73.33%
Epoch [17/30] Train Loss: 0.3946 Test Loss: 0.8523 Test Acc: 73.33%
Trainable Parameters: 612,714

Resnet 18(Dropout .5 was used)

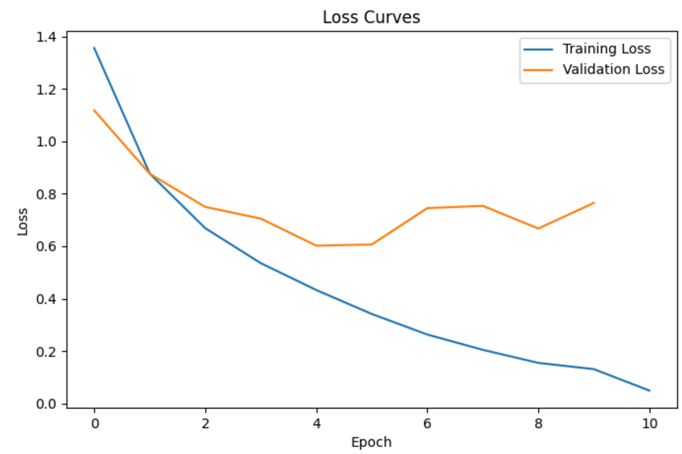


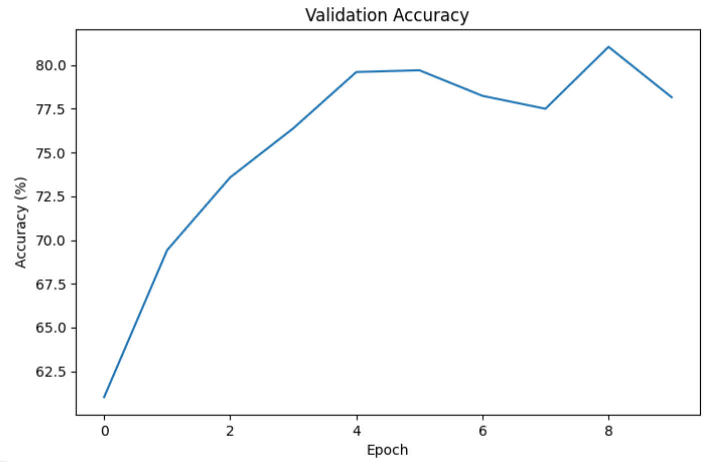


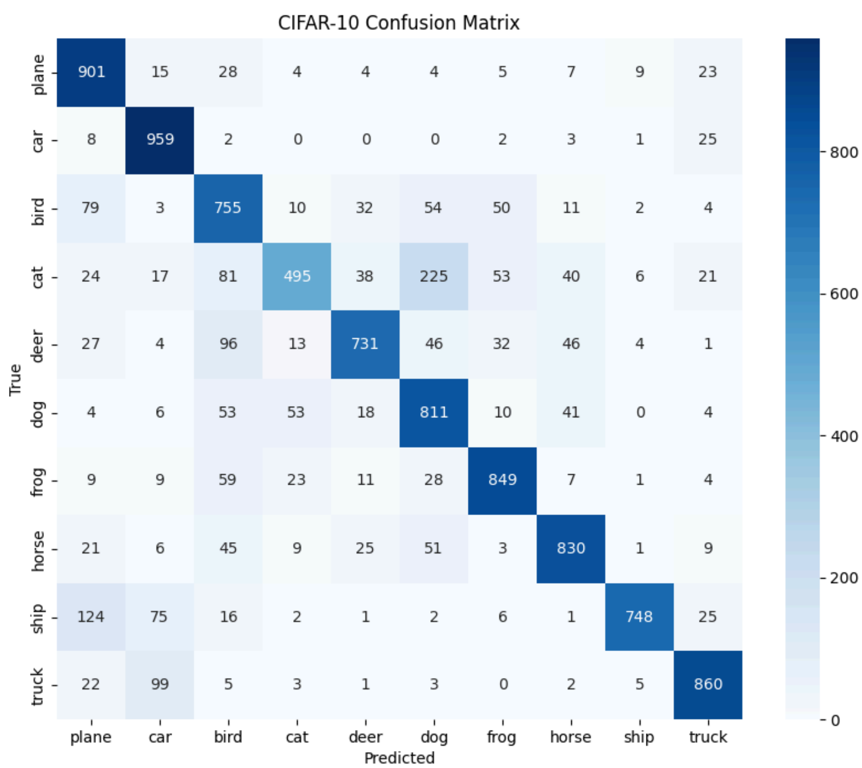


Total Training Time: 1029.65 sec
Average Epoch Time: 61.06 sec
Seed 1: 86.09%
Epoch [17/30] Train Loss: 0.0039
Test Loss: 0.4571 Test Acc:
86.09%
Trainable Parameters: 11,173,962

Resnet 11(Drop out .5 was used)







Total Training Time: 426.82 sec
Average Epoch Time: 39.06 sec
Seed 1: 79.39%
Epoch [11/30] Train Loss: 0.0495 Test Loss: 0.6087 Test Acc: 79.39%
Trainable Parameters: 4,903,242